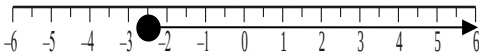
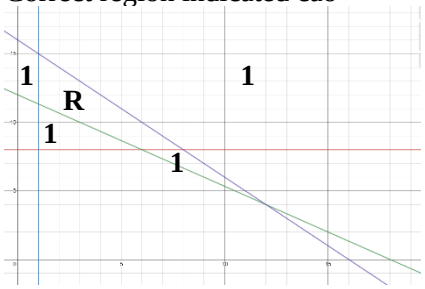
0580_w22_ms_43_Q6

Question	Answer	Marks	Partial Marks
6(a)(i)	38	**2**	**M1** for $5 \times 3^2 - 7$ oe
6(a)(ii)	$[\pm]\sqrt{\frac{P+7}{5}}$ oe final answer	**3**	**M1** for $P + 7 = 5k^2$ or $\frac{P}{5} = k^2 - \frac{7}{5}$ **M1** for $k^2 =$ ……. FT *their* first step **M1** for square root to final answer Max M2 for incorrect answer
6(b)(i)	$x \geqslant -2.5$ final answer	**2**	**M1** for $-4x \leqslant 7 + 3$ or better
6(b)(ii)		**1**	**FT** *their* inequality in **(b)(i)**
6(c)(i)	$x = 2$ broken line	**B1**	
	$y = 32 - x$ solid line	**B1**	
	$2x + 3y = 72$ solid line	**B2**	**B1** for line passing through (0, 24) or (36, 0)
	Correct region indicated cao 	**B2**	**B1** for region satisfying 3 of the inequalities
6(c)(ii)	(16, 16)	**2**	**M1** for substitution into $2x + y$ for any integer point in *their* region